

**ATBM6162/
6062C**1x1 802.11b/g/n/ax
Wi-Fi 芯片**APPLICATION NOTE****ATBM BLE 认证需要用的命令****Table of contents**

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ATBM BLE 认证测试指令

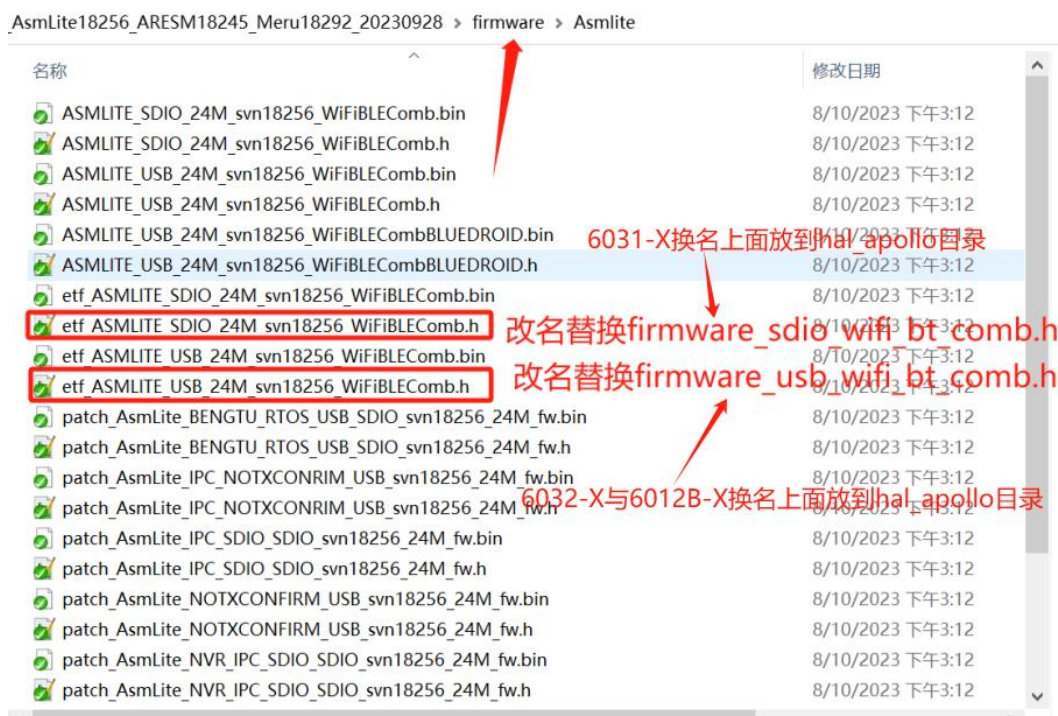
作者	说明	版本	日期
Fuzhenzhou	初始版本	v1.0	20250208
Wangxixi	修改格式	v1.1	20250214

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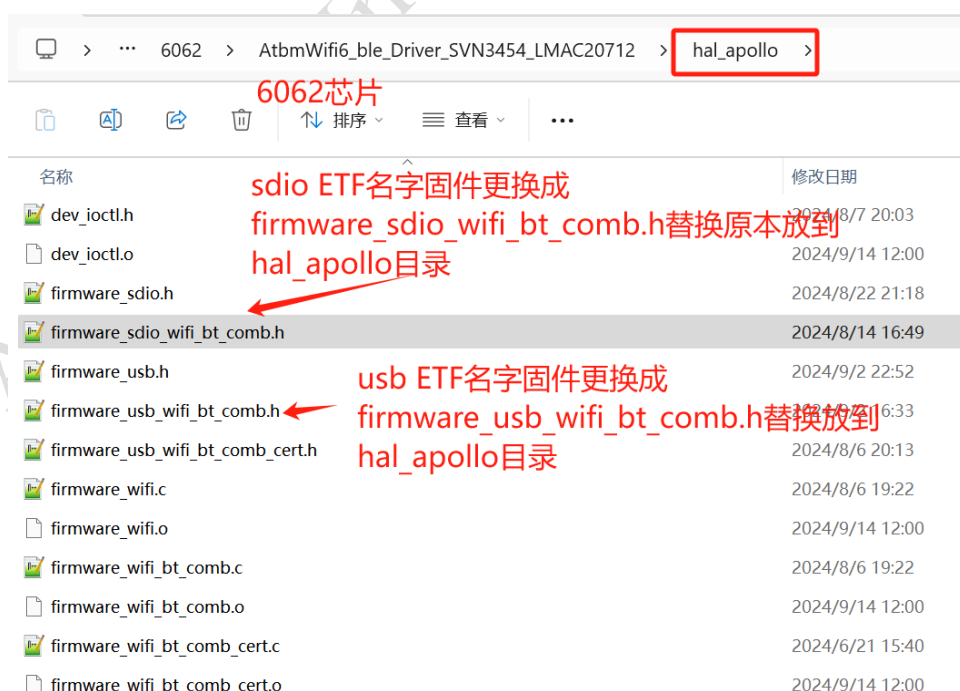
1 Linux 长电系列芯片固件替换方法（6461 不需要看）

其中 ATBM6062、ATBM6031、ATBM6032、ATBM6132、ATBM6012B 芯片需要在 firmware 目录下把下图文件名字带有 etf 开头.h 文件改名后放在 hal_apollo 目录下替换编译生成专门测试的 BLE 固件。

1) ATBM6031、ATBM6032、ATBM6012B 芯片 BLE 固件的替换方法如下图所示：



2) ATBM6062 芯片 BLE 固件的替换方法如下图所示：



3) ATBM6132 芯片 BLE 固件的替换方法如下图所示：



而其他芯片请对应下面型号命名：

芯片型号	WiFi+BLE comb firmware
ATBM6031-X	firmware_sdio_wifi_bt_comb.h
ATBM6032-X	firmware_usb_wifi_bt_comb.h
ATBM6132	firmware_usb_mercurius_wifi_bt_comb.h
ATBM6132C	firmware_sdio_wifi_bt_comb_ocea.h firmware_usb_wifi_bt_comb_ocea.h
ATBM6062	firmware_sdio_wifi_bt_comb_cronus.h firmware_usb_wifi_bt_comb_cronus.h
ATBM6062C	firmware_sdio_wifi_bt_comb_clite.h firmware_usb_wifi_bt_comb_clite.h
ATBM6162	firmware_sdio_wifi_bt_comb_ocea.h firmware_usb_wifi_bt_comb_ocea.h

更换完后请确认版本号跟替换 **etf** 固件版本号是否一致：

```
618556] [atbm_log]:apollio will wsm init done.
618556] Input buffers: 18 x 1648 bytes
618556] Hardware: 7.0
618556] WSM firmware [=MODEM=RF=Ares_AX Feb 23 2024 18:03:05], ver: 19624, build: 4144, api
618558] [atbm_log]:EFUSE(8) [0]
618560] [atbm_log]:EFUSE(I) [0]
618563] [atbm_log]:EFUSE(B) [0]
618566] [atbm_log]:CAPABILITIES_ATBM_PRIVATE_IE [0]
618569] [atbm_log]:CAPABILITIES_NVR_IPC [0]
618572] [atbm_log]:CAPABILITIES_NO_CONFIRM [1]
618575] [atbm_log]:CAPABILITIES_SDIO_PATCH [0]
618578] [atbm_log]:CAPABILITIES_NO_BACKOFF [0]
```

Ble log:

```
root@TinaLinux:/# atbm_ble
=====>>atbm_ble_start>>>
nimble_port_init++
nimble_host_task
FD_CLOEXEC set
ble_coexist_start--
hif_ioctl_loop
get_command_func
ble_hs_sync:333
ble_hs_hci_cmd_tx:352
ble_smart_cfg_startup!11

[ATBM LOG]set default tx power(0)
ble_hs_hci_cmd_tx:352
error setting advertisement data; rc=513
pub addr 98:a8:29:ad:9a:b5
```

替换成功启动ble广播就会失败

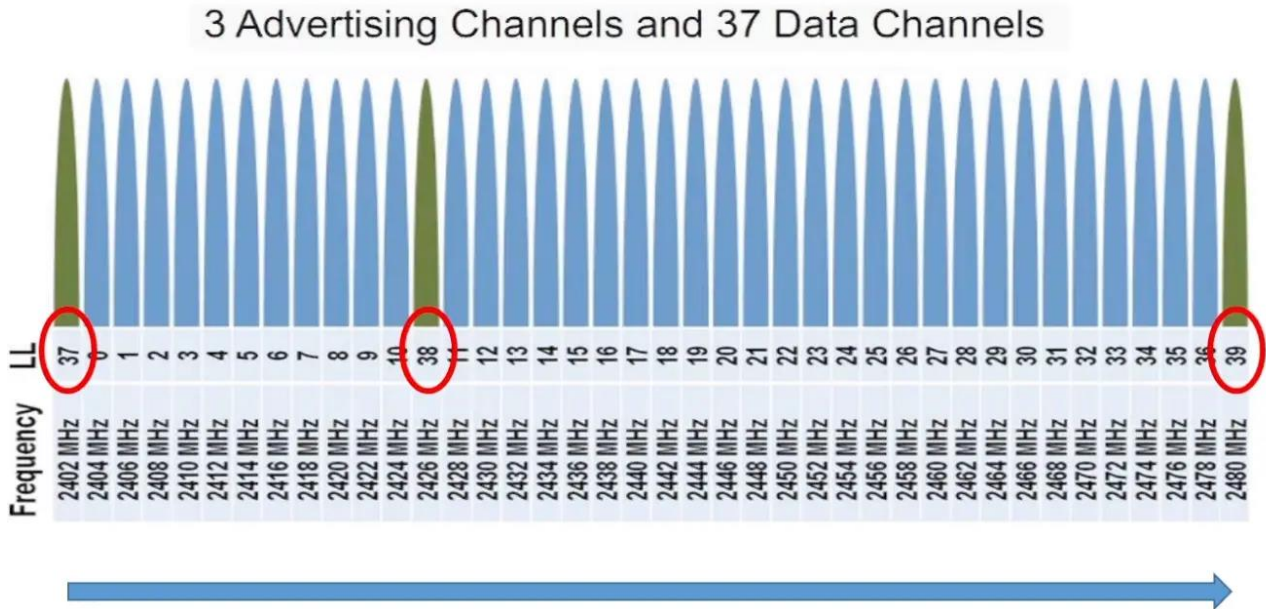
1.1 加载驱动

```
insmod ATBM606x_wifi_usb.ko wifi_bt_comb=1 //请根据实际驱动名字
```

2 Linux 长电系列非信令测试（6461 不需要看）

```
./nimble_linux & //需要把 BLE 协议栈运行起来才能测试
```

3 BLE RF 信道介绍



4 测试指令

注：在第 4 章节 6461 芯片所有测试项不需输入 `atbm_cli`，请直接输入 `btshell XXXX`

4.1 开启 DTM TX 测试

命令	<code>btshell dtm-tx chan=<channel> len=<length> payload=<packet_Payload> phyMode=<PHY Mode></code>
参数说明	<channel>: 信道 $channel = (F - 2402) / 2$，用十六进制表示，是 RF channel 不是 channel index 比如 $channel = 2$ 反推 RF channel 为 2406 公式: $(2406 - 2402) / 2 = 2$ <length>: Length in bytes of payload data in each packet (0x00-0xff) <packet_Payload>: 0x00 PRBS9 0x01 Repeated '11110000' 0x02 Repeated '10101010' 0x03 PRBS15 0x04 Repeated '11111111' 0x05 Repeated '00000000' 0x06 Repeated '00001111' 0x07 Repeated '01010101' <PHY Mode> : 1 1M PHY 2 2M PHY

	3 S8 Code PHY 4 S2 Code PHY
功能描述	开始 Direct Test TX Mode
返回值	OK or Fail
Linux 长供电例子	atbm_cli btshell dtm-tx chan=2 len=27 payload=1 phyMode=1
6461 低功耗例子	btshell dtm-tx chan=2 len=27 payload=1 phyMode=1

4.2 停止 DTM TX 测试

命令	btshell dtm-tx stop
参数说明	无
功能描述	结束 Direct Test TX Mode
返回值	OK or Fail
Linux 长供电例子	atbm_cli btshell dtm-tx stop
6461 低功耗例子	btshell dtm-tx stop

4.3 启动 DTM RX 测试

命令	btshell dtm-rx chan=<channel> phyMode=<PHY Mode>
参数说明	<channel> : 信道 $channel = (F - 2402) / 2$, 用十六进制表示, 是 RF channel 不是 channel index 比如 channel=2 反推 RF channel 为 2406 公式: $(2406 - 2402) / 2 = 2$ <PHY Mode>: 1 1M PHY 2 2M PHY 3 Receiver set to use the LE Coded PHY
功能描述	开始 Direct Test RX Mode 1. ifconfig wlan0 up 2. 需要打开 lmac 打印 (iwpriv wlan0 fwdbg 1) 需要调用 btshell dtm-rx stop 指令后查看字符串 "ll_dtm_end:" 收到多少包
返回值	OK or Fail
Linux 长供电例子	atbm_cli btshell dtm-rx chan=2 phyMode=1
6461 低功耗例子	btshell dtm-rx chan=2 phyMode=1

4.4 停止 DTM TX 测试

命令	btshell dtm-tx stop
参数说明	无
功能描述	结束 Direct Test RX Mode
返回值	OK or Fail
Linux 长供电例子	atbm_cli btshell dtm-rx stop

6461 低功耗例子	btshell dtm-rx stop
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4.5 启动单载波测试

命令	btshell tone-tx chan=<channel> power=<level>
参数说明	<channel> : 信道 $channel = (F - 2402) / 2$, 用十六进制表示, 是 RF channel 不是 channel index 比如 channel=2 反推 RF channel 为 2406 公式: $(2406 - 2402) / 2 = 2$ <level> : 0 或者 1 等级
功能描述	启动单载波测试
返回值	OK or Fail
Linux 长供电例子	atbm_cli btshell tone-tx chan=0 power=0
6461 低功耗例子	btshell tone-tx chan=0 power=0

4.6 停止单载波测试

命令	btshell tone-tx stop
参数说明	无
功能描述	停止单载波测试
返回值	OK or Fail
Linux 长供电例子	atbm_cli btshell tone-tx stop
6461 低功耗例子	btshell tone-tx stop

4.7 修改发射功率

命令	btshell pwr-set tx_pwr=<level>
参数说明	tx_pwr=<level> 范围值(0 ~ 15)在启动发包之前设置功率
功能描述	修改 ble 发射功率
返回值	OK or Fail
Linux 长供电例子	atbm_cli btshell pwr-set tx_pwr=0
6461 低功耗例子	btshell pwr-set tx_pwr=0

5 Linux 长电系列 DTM 信令测试 (6461 不需要看)

- 1) 不需要执行 `./nimble_linux &`
- 2) 需要找出芯片 uart 飞线出来:

芯片 uart 默认有 log 输出，也可以输入按回车看看有没有反应

```

]dsss1,0,clear4,Err0,unsup0,pwrup0,lost0
]num[tx1,rx0],idle[236,0]
]ofdm[229,19],dsss[1,0],over0,sound0
]AckTimeout[189,31,63351],ChLen[0,205,24,0,0]
]ppm-1,cnt127
]gain26
]rssi-41,gain28->26,small0
]CPU: 1%
]Temp68,ccast1,psm50001,inig26->28,nodBm-57,sig0
]dsss0,1,clear1,Err0,unsup0,pwrup0,lost0
]num[tx1,rx0],idle[1024,0]
]ofdm[0,0],dsss[0,0],over0,sound0
]AckTimeout[60,10,0],ChLen[0,52,7,0,0]
]gain26
]rssi-41,gain26->26,small0
]Temp68,ccast1,psm50001,inig26->34,nodBm-68,sig0
]dsss0,0,clear1,Err0,unsup0,pwrup0,lost0
]num[tx1,rx0],idle[1024,0]
]ofdm[0,0],dsss[0,0],over0,sound0
]AckTimeout[171,10,1],ChLen[0,14,1,0,0]
]ppm-1,cnt127
]CPU: 1%
]Temp68,ccast0,psm5,inig26->0,nodBm-97,sig0
]dsss0,0,clear0,Err0,unsup0,pwrup0,lost0
]num[tx0,rx0],idle[0,0]
]ofdm[0,0],dsss[0,0],over0,sound0
]AckTimeout[162,19,0],ChLen[0,45,4,0,0]
]gain30
]rssi-44,gain26->30,small0
]Temp68,ccast1,psm50001,inig30->40,nodBm-73,sig0
]dsss0,0,clear15,Err0,unsup0,pwrup0,lost0
]num[tx1,rx0],idle[796,0]
]ofdm[0,0],dsss[1,0],over0,sound0
]AckTimeout[208,44,3756],ChLen[0,47,7,0,0]
]gain30
]rssi-44,gain30->30,small0
]ppm-1,cnt127
]CPU: 1%

```

- 3) 需要切换成 UART HCI 测试模式，首先要将 wlan0 给启动起来:

```
ifconfig wlan0 up wlan0
```

- 4) 启动成功之后，通过如下命令可以切换 UART HCI 模式，还是普通模式

```
iwpriv wlan0 fwcmd set_ble_uart,1 //切换到 UART HCI 模式
```

```
iwpriv wlan0 fwcmd set_ble_uart,0 //切换到普通 HOST 下发 DTM 模式
```

- 5) 串口波特率 115200,8,1,无校验，无 RTS/CTS 当切换到 UART HCI 模式之后，我们的串口无法再输入指令，此时给我们的串口发送

```
Send COM: 01 03 0C 00 (Reset)
```

- 6) 可以看到我们的串口的回复，即进入 UART HCI 测试模式成功，可以接测试仪器进行测试

```
Receive COM: 04 0E 04 01 03 0C 00 (7 bytes)
```

- 7) 最后把芯片串口接到测试仪器 CMW500 测试就可以了。

6 6461 DTM 信令模式测试

- 1) 进入 CMW500 DTM 测试模式

- 2) 在我们的 SDK 启动的情况下，关闭 WIFI 扫描连接等动作，让 WIFI 完全停下来

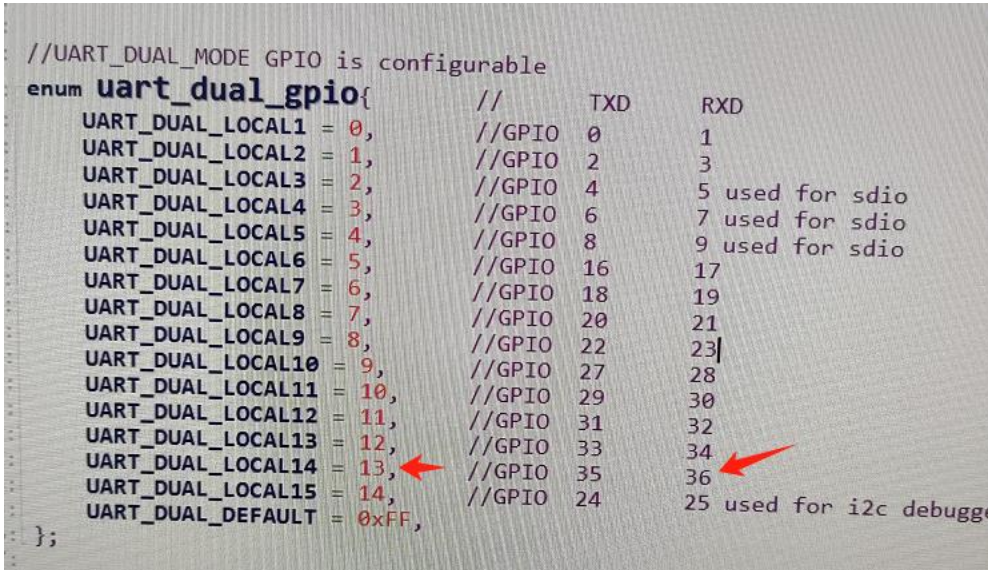
AT+WIFI_DISCONNECT

- 3) 关闭 BLE 广播扫描连接状态后然后输入命令:

btshell advertise stop

- 4) 配置 ble HCI uart 引脚映射为第 13 组 GPIO35 和 36

btshell hci-uart-test start gpio_local=13



```
//UART_DUAL_MODE GPIO is configurable
enum uart_dual_gpio{
    UART_DUAL_LOCAL1 = 0, //GPIO 0 TXD 1
    UART_DUAL_LOCAL2 = 1, //GPIO 2 TXD 3
    UART_DUAL_LOCAL3 = 2, //GPIO 4 TXD 5 used for sdio
    UART_DUAL_LOCAL4 = 3, //GPIO 6 TXD 7 used for sdio
    UART_DUAL_LOCAL5 = 4, //GPIO 8 TXD 9 used for sdio
    UART_DUAL_LOCAL6 = 5, //GPIO 16 TXD 17
    UART_DUAL_LOCAL7 = 6, //GPIO 18 TXD 19
    UART_DUAL_LOCAL8 = 7, //GPIO 20 TXD 21
    UART_DUAL_LOCAL9 = 8, //GPIO 22 TXD 23
    UART_DUAL_LOCAL10 = 9, //GPIO 27 TXD 28
    UART_DUAL_LOCAL11 = 10, //GPIO 29 TXD 30
    UART_DUAL_LOCAL12 = 11, //GPIO 31 TXD 32
    UART_DUAL_LOCAL13 = 12, //GPIO 33 TXD 34
    UART_DUAL_LOCAL14 = 13, //GPIO 35 TXD 36
    UART_DUAL_LOCAL15 = 14, //GPIO 24 TXD 25 used for i2c debugge
    UART_DUAL_DEFAULT = 0xFF,
};
```

- 5) 此时就进入了 cmw500 的测试模式，cmw500 仪器可以通过对应的串口连接我们的设备给我们发送 DTM 命令进行测试，波特率 115200，连接的串口取决于 gpio.local 的串口映射，默认映射是第 13 组，即 GPIO35 36。



CONTACT INFORMATION

AltoBeam (China) Inc.

Address: B808, Tsinghua Tongfang Hi-Tech Plaza, Haidian, Beijing, China 100083

Tel: (8610) 6270 1811

Fax: (8610) 6270 1830

Website: www.altobeam.com

Email: support@altobeam.com

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